Department of Microbiology
Alagappa University

[Accredited with A+ Grade by NAAC (CGPA: 3.64) in Third Cycle, Graded as Category-I University and Granted Autonomy by MHRD-UGC]

Karaikudi

M.Phil Microbiology

(Regular Programme)

2019-20 onwards

Regulation and Syllabus

(For those admitted in academic year 2019-20 onwards)

1. **a. Programme general objectives**

Microbiology has become increasingly important to human society. It has emerged as one of the most important branches of life sciences. As microbes practically affect all activities of our life like, food, clothing, shelter, health hygiene etc., so microbiology has made vast progressive strides in all these fields in little less than a century to improve the quality of our life.

Hence our task is to introduce the M.Phil. Programme in Microbiology to educate the postgraduate students in the captivating fields. Rigorous and comprehensive in approach, this syllabus presents essential contents in a detailed, clear and direct way. This programme is offered under Choice Based Credit System (CBCS). The CBCS enables the students to select variety courses as per his interest and requirement. The programme is structured in such a way to impart more knowledge in science, in particular in Microbiology.

b. Programme specific objectives

1. To acquire knowledge on the advances of microbiology
2. To make the students understanding on fundamental interaction of the microbes with other biological and non biological elements.
3. To understand the rationale in the field of applied microbiology
4. To enable the students technically sound in the microbial techniques
5. To prepare the students to do research on the recent trends in Microbiology
6. To make them expert in the field of applied Microbiology

c. <u>Programme outcome</u>

After successful completion of this course the student will be able to,

1. Become expertise in the field of microbiology in research aspect.
2. Will receive elaborate knowledge in the field of Microbiology, Biochemistry, Microbial genetics, Molecular biology, Food, Agricultural, Environmental, Medical and applied Microbiology
3. Will be capable of carrying out any Microbiology related tasks in Industries, Medical labs, Research labs and etc.

A postgraduate degree in microbiology prepares the students for a career in research. It is important to start to think about your plans after your master's or PhD so that you can make the most of the ever-increasing options available to postgraduate students.

II. Eligibility for Admission

A candidate who has passed M.Sc., Degree in any life Sciences discipline as main course of study of any university accepted by the syndicate as equivalent thereto, subject to such condition as may be prescribed therefore shall be permitted to appear and qualify for the M.Phil degree in Microbiology of this University after a course of study of one academic year.

III. Duration of the Course

The course for the degree of M.Phil Microbiology shall consist of one academic year divided in to two semesters. Each Semester consist of 90 working days.

IV. Course of Study: M.Phil Microbiology

V. Examinations

The examination shall be three hours duration to each course at the end of each semester. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

For Subject 517102 examination shall be held at end of the first semester. It will be conducted by the department in a single session for all the students. Guide has to set the question paper confidentially and hand over to the HOD well in advance. The marks obtained by the candidate along with syllabus, question paper and valued answer scripts shall be sent to Controller of Examinations.

At the end of second semester, viva-voce will be conducted on the basis of the Dissertation report submitted by the student. Viva-Voce will be conducted jointly with the following members - H.O.D – Member of the Viva Examination, Research Supervisor and External examiner from outside (Appointed by University selected from the panel of four members).

VI. Question Paper Pattern – Theory

M.Phil Microbiology
Course Code: Course Title
(2019-20 onwards)

Time: 3 Hours **Max. Marks: 75**

Answer all questions either (a) or (b) **(5 x 15 = 75 marks)**

1. (a) or (b)
2. (a) or (b)
3. (a) or (b)
4. (a) or (b)
5. (a) or (b)

Each Question is represented from the same unit.

VII. Dissertation Work

Dissertation Work	–	150 marks
Viva-Voce	–	50 marks
Total	–	200 marks

(a) Plan of Work:

The student should prepare plan of work under the supervision of guide for the dissertation, get the approval of the guide and should be submitted to the university during the second semester of their study. The duration of the dissertation research shall be a minimum of six months in the second semester.

(b) Project Work outside the Department:

In case the student stays away for work from the Department for more than one month, specific approval of the university should be obtained.

(c) No. of copies/distribution of project work:

The students should prepare three copies of dissertation and submit the same for the evaluation by Examiners. After evaluation one copy is to be retained in the Department library and one copy is to be submitted to the guide and one copy can be held by the student.

(d) Format to be followed:

The format/certificate for dissertation to be submitted by the students is given below:

Format for the preparation of project work:

(a) Title page
(b) Bonafide Certificate
(c) Acknowledgement
(d) Table of contents

CONTENTS

Chapter No.	TITLE	Page No.
1.	Introduction	
2	Review of Literature	
3.	Materials and Methods	
4.	Results	
5.	Discussion	
6.	Summary	
7.	References	

Format of the Title Page:

TITLE OF THE PROJECT

A Dissertation Submitted to the Alagappa University, Karaikudi -630 003 in Partial Fulfillment of the Requirement for the Award of Degree of

MASTER OF PHILOSHOPHY IN MICROBIOLOGY

By

Students Name

Register Number

Supervisor:

University Emblem
Department of Microbiology
Alagappa University
Karaikudi – 630 003
Month and Year

Format of Declaration of the Candidate:

Name and class of the student

DECLARATION

I hereby declare that the Project entitled _____ submitted to Alagappa University for the award of the degree of Master of Philosophy in Microbiology is my original work and that it has not been previously formed the basis for the award of any degree, diploma/associate ship or any other similar title of any other University or Institution.

Signature of the Student

Format of the Certificate:

CERTIFICATE

This is to certify that the project entitled ----------submitted in partial fulfillment of the requirement of the degree of Master of Philosophy in Microbiology to the Alagappa University, Karaikudi is a record of bonafide research work carried out by ----------under my supervision and guidance and that no part of the project has been submitted for the award of any degree or diploma.

HOD Endorsement Signature of Guide

Place:

Date:

VIII. Classification of Successful Candidates

P.G. Programme

MARKS	GRADE POINT	CGPA	LETTER GRADE	Description
96 and above	10	9.51 and above	S+	First Class - Exemplary
91 – 95	9.5	9.01 – 9.50	S	
86 – 90	9.0	8.51 – 9.00	D++	First Class – Distinction
81 – 85	8.5	8.01 – 8.50	D+	
76 – 80	8.0	7.51 – 8.00	D	
71 – 75	7.5	7.01 – 7.50	A++	First Class
66 – 70	7.0	6.51 – 7.00	A+	
61 – 65	6.5	6.01 – 6.50	A	
56 – 60	6.0	5.51 – 6.00	B	Second Class
50 – 55	5.5	5.00 – 5.50	C	
Below 50	-	Below 5.00	F	Fail
			AA	Absent

Passing Minimum: 50%, P: Pass, ESE: End Semester Examination, CIA: Continuous Internal Assessment

$$GPA = \frac{\Sigma\,(CDT * GPT)}{\Sigma CDT}$$

Where: CDT = No. of credits of major allied and elective courses

GPT = Grade Point (obtained by dividing the percentage of marks scored by 10)

Note: Extra Grade Points and Marks are not considered for GPA and Total Marks Calculations.

CGY – Category CDT – Credit GRD – Grade GPT – Grade Point GPA – Grade Point Average CGPA – Cumulative Grade Point Average	**<u>Category (CGY)</u>** 1. Theory (Core) 2. Practical (Core) 3. Inter-disciplinary – Theory 4. Inter-disciplinary – Practical 5. Elective / Optional 6. Comprehensive Viva / Seminar 7. Extension Programmes 8. Project and Viva – Voce

IX. Syllabus

IX. Syllabus - M.Phil Microbiology

Semester – I					
Course Code	Course Title	Credits	Max Marks I	Max Marks E	Total
517101	CC I - Research Methodology	4	25	75	100
517102	CC II - Topic of Research	4	25	75	100
517103	EC I - General Skills In Science	4	25	75	100
	Total	12	75	225	300
Semester – II					
517201	CC III - Applied Microbiology	4	25	75	100
517202	CC IV - Project Work	8	200 (Thesis 150 + Viva Voce 50)		200
	Total	12	25	275	300
	Total for All Semesters	24	100	500	600

Semester - I		
Course code: 517101	**Research Methodology**	**Credits:4**
Objectives	➢ To inculcate knowledge on research methodology ➢ To familiarize in techniques used in biology ➢ To know the principles of various instruments. .	
Unit -I	**Literature collection:** Definition, basic concepts, objectives, significance and techniques of research, finding research materials – literature survey, compiling records. Scientific documents – research paper, review paper, book reviews, theses, conference and project reports.	
Unit-II	**Basic analytical techniques:** pH meter- determination of pH –buffers – principle –operation and description of the apparatus. Dissociation constant of acids – titration of weak acid with a strong base. Colorimetry - ultraviolet - visible spectrophotometry - principles, instrumentation and applications, Fluorescence spectrophotometry, FTIR and XRD- , instrumentation and applications.	
Unit III	**Chromatography**: paper, column, thin layer, gas, liquid and ion exchange chromatography – Principle - description– applications of various chromatography. Electrophoresis – Principle, method and application of acrylamide and agarose gel electrophoresis. Two dimensional gel electrophoresis (IEF), DGGE, trflp and RAPD.	
Unit IV	**Basic principles of microscope:** Basic principles and sedimentation - types of centrifuge – density gradient and ultra-centrifuge – applications. Basic principles of microscope – compound, electron, phase contrast, fluorescent, microscopy, TEM, SEM, AFM – applications.	
Unit V	**Spreadsheet tools** : Introduction to spreadsheet applications, features, Using formulas and functions, Data storing, Features for Statistical data analysis, Generating charts / graph and other features, Tools – Microsoft Excel or similar. Presentation tools: Introduction, features and functions, Presentation of Power Point Presentation, Customizing presentation, Showing presentation, Tools – Microsoft Power Point or Similar. Writing of Synopsis and dissertation and thesis.	

Reference and Textbooks:-

1. Bajpai, S. (2014). *Biological Instrumentation and Methodology*: (*Tools and Techniques of Biology*) (Revised). New Delhi: Chand & Company Ltd.
2. Devlin, T.M. (1982). Devlin: *Textbook of Biochemistry – With Clinical Correlations*, John Wiley & Sons.
3. Gurumani, N. (2016). *Research Methodology for Biological Sciences* (1st ed). Chennai: MJP Publishers, A unit of Tamilnadu Book House.
4. Lederberg, J. (2000). *Encyclopedia of Microbiology* Volume 4 (2nd ed). Academic Press.
5. Lehninger A.L. (2015). *Biochemistry*, Kalyani Publishers
6. Palanivelu, P. (2009). *Analytical biochemistry and separation techniques – A Laboratory Manual* (4th ed). Madurai: Twenty first Century Publications.
7. Plummer, D.T. (2003). *An Introduction of practical biochemistry* (3rd ed). New Delhi: Tata McGraw Hill Publishing Company Ltd.
8. Prave, P., Faust, U., Sittig, W. and Sakatsch, D.A. (2004). *Fundamentals of Biotechnology*. (1st ed). India: Panima Publishing Corporation.
9. Sawhney, S.K. & Singh, R., (2005). *Introductory Practical Biochemistry* (2nd ed). Alpha Science International Ltd.
10. Verma, A.S., Surajit, D & Anchal, S. (2014). *Laboratory Manual for Biotechnology*. New Delhi: S. Chand and Company Ltd.

Outcomes	➢ Knowledge on research methodology ➢ Basic knowledge on different instruments ➢ Ideas on research review

Name of the Course Teacher: T. Kavitha

Semester I

Course II	**: Topic of Research**	**Maximum Marks: 100**
Code	**: 517102**	**Internal Marks : 25**
Credit	**: 4**	**External Marks : 75**

(It is a guide's paper. Sub title and syllabus have to be given by the guide as per the guidelines given below)

Unit I – FUNDAMENTAL CONCEPTS

Identification of research problem. Fundamental information on the research topic selected. Significance of the research topic chosen.

(A minimum of 20 pages of materials to be prepared)

Unit II – LITERATURE SURVEY I

Literature survey on the research problem. Use of text books, reference materials, journals, internet and specialized softwares for literature collection. The period of literature search restricted upto the year 2009.

(A minimum of 6 research articles may be given)

Unit III – LITERATURE SURVEY II

In-depth study of published papers, seminar / conference proceedings on the identified research problem. The period of literature search extended from the year 2010 to the current date. National and International status on the research topic.

(A minimum of 6 research articles may be given)

Unit IV – PREPARATORY METHODS

Procurement or synthesis of chemicals needed for the research problem. Database, storage and handling of specified chemicals. Preparatory methods to execute the experimental works. Details on the principle, instrumentation and working of instruments identified for the problem.

(A minimum of 20 pages of materials to be prepared)

Unit V – CHARACTERIZATION TECHNIQUES

Types of instrumental techniques used for the identified research problem. Characterization of research materials, interpretation of results and preparation of manuscript. Writing of Research report / thesis. Presentation of research output in scientific seminar / conference.

(A minimum of 30 pages of materials to be prepared)

Semester - I		
Course code: 517103	**General Skills in Science**	**Credits:4**

Objectives	➢ To learn about computers ➢ To know the applications of computer ➢ To Learn about the Communication skills and Pedagogical skill
Unit –I	**Introduction to Computers:** Computer Hardware: Input devices and media-magnetic device and media-output devices and media-storage device and media-computer architecture – system software: types, operating system and translators – Application software: types of language – application packages – integrated software – Introduction to operating system – Working with windows and office programs – Internet, Website and Email
Unit-II	**Computer operating skills :** Starting a program and opening a document – saving and naming the document – create file and folders – deleting and un-deleting a document – closing a document – renaming and moving a document – finding a document – MS office: Word, Excel, Access, Power point, outlook and integrated office applications – C programming – Principles, classes and structure of C++ programming
Unit III	**Communication skills in English :** Understanding communication – greeting and introducing – making requests – asking for and giving permission – offering help – giving instruction and directions art of small talk participating in conversation – making a short formal speech – Describing the people, place, events and things. Telephone skill: understanding, handling calls, leaving message and making request. Written communication: report writing, note making – career skills: curriculum vita and cover letter – Facing an interview and presentation skills – academic listening
Unit IV	**Pedagogical skill for science teachers:** Science Teacher: Qualification, teacher competencies and professional growth. Theory and models of curriculum development: Concept and Technical scientific models of curriculum development – planning a science library – Handling of practical classes. Educational technology and classroom pedagogy: educational Technology – Concept, Emerging technologies – New technologies on methodology of teaching, learning experiences and curriculum development. Micro-teaching: Meaning, teaching, skill of stimulus variation, questioning, explanation, reacting, linking and benefits
Unit V	**Practical training:** Preparation of charts and models for handling classes of science teacher – Creating management documents e.g. Curriculum Plan, Time Table scheduling, Evaluation – Strategies etc – Learning to write and draw on the blackboard – Preparation of over head projector presentations – Preparation of power point / LCD presentations – Preparation of micro-teaching skills – Preparation of teaching materials – seminar classes for PG students – Preparation of album

Reference and Textbooks:-

1. Benny Raphael, F.C.Smith. (2003). *Fundamentals of computer-aided engineering*, John wiley & sons.
2. Curtis Frye. (2004). *Microsoft office Excel 2003 step by step.* Microsoft press.
3. Greg Harvey. (2006). *Microsoft office Excel 2007 for dummies*. For Dummies.
4. Guy Hart-D Eavis. (2007). *How to do everything with Microsoft office word 2007*. Mac Graw-Hill professional.
5. Louis Rosenblatt. (2010). *Rethinking the Way We Teach Science: The Interplay of Content, Pedagogy and the Nature of Science*. Taylor & Francis.
6. *National Science Education Standards: observe, interact, change, learn, manual.* (1996). National research council, National Academics Press.
7. Raja Dan, Durga., Kash. U. (1982). *A Study of the International Influences in Functional Content on curriculum Programme.*
8. Taba, Hilda. (1962). *Curriculum Development, Theory and Practice,* Harcouort Brace and World Inc.
9. Vijaya Kohli. (1992). *How to teach science: a treatise on methodology of teaching physics, chemistry and biology,* Vivek Publishers.
10. W. Joseph, Habraken. (2004). Microsoft office 2003, All in one, Que publishing.

Outcomes	➢ Better understanding importance of computer and its application ➢ Get information about Pedagogical skills ➢ Enable them to have good communication in English

Name of the Course Teacher: Dr. V. Balasubramanian

Semester - II		
Course code:517201	**Applied Microbiology**	**Credits: 4**

Objectives	➢ To acquire knowledge on food product analysis ➢ To enable them to know about preservation of pharmaceutical products ➢ Learn to assess the microbial quality of marine foods
Unit -I	**Mineral water industry:** Stages of mineral water production. Analysis of water quality – pH, salinity, alkalinity, dissolved oxygen, carbonates, nitrate, silicate, phosphate, COD and BOD. Determination of microbial load in water : **Faecal indicator organisms -** coliform bacteria, faecal enterococci, *Clostridium perfringens*, yeast, mould and sulphide reducing anaerobes, viruses and bacteriophages, fungi and yeasts, protozoa and helminths. **Methods of mineral water quality assessment** – MPN test, membrane filtration technique.
Unit-II	**Preservation of pharmaceutical Products:** Chemical preservatives – raw materials – equipment – role of preservatives. **Finished product tests** – microbial enumeration test, tests for specified microorganisms. **Sterility testing** – antimicrobial effectiveness testing. Sterility assurance – biological indicators, sterilization validation process. Microbial risk assessment through HACCP plan.
Unit III	**Endotoxin test methods -** gel clot assay, turbidometeric assay and chromogenic methods. **Biological assays -** vitamin assay, antibiotic assay and mycoplasma testing. **Endotoxin activity** – risk assessment in parenterals manufacture – pyrogen test – depyrogenation methods.
Unit IV	**Rapid methods for detection of microorganisms in food** – conventional and automated. Application of light pulse technology – principles of light pulse generation, mode of action, equipments, application of light pulses, effect of light pulses on foods and microorganisms, advantage and limitation of light pulse treatment. Quality control in fruits and vegetable processing. Risk assessment in food industry – physical, chemical and biological hazards.
Unit V	**Assessment of microbial quality of marine foods:** Conventional and recent development methods – flow cytometry, ATP estimation, radiometric, reflective colorimetry, LAL test, immunoassay, DNA based and microarray methods. Application of additives in food.Food safety and standard act for adulteration.Significance of barcode and its uses in food industry.

Reference and Textbooks:-

1. Aneja, K.N. (2018). *Modern Food Microbiology*, Medtec Publisher.
2. Bhatnagar, R. (2017). *Food Microbiology*, Crescent Publishing Corporation
3. Carlberg, D.M. (2004). *Cleanroom Microbiology for the Non-Microbiologist* (2nd ed). CRC Press.
4. Clontz, L. (2008). *Microbial Limit and Bioburden Tests: Validation Approaches and Global Requirements* (2nd ed). CRC Press.
5. Plummer, D.T. (2003). *An Introduction of practical biochemistry* (3rd ed). New Delhi: Tata McGraw Hill Publishing Company Ltd.
6. Prince, R. (2008). *Microbiology in Pharmaceutical Manufacturing* Volume 1 (2nd ed, Revised and Expanded) Parenteral Drug Association and Davis Healthcare International Publishing.
7. Trivedi, R. (2016). *Practical Mannual in Microbial Physiology and Industrial Microbiology*. New Delhi SSDN Publishers.
8. Trivedy, R.K., Goel, P.K. and Trishal, C.L. (1987). *Practical methods in Ecology and Environmental science*, Environmental publishers.
9. Williams, K.L. (2004). *Microbial Contamination Control in Parenteral Manufacturing.* New York: Marcel Dekker, Inc.
10. Williams, K.L. (2007). *Endotoxins: Pyrogens, LAL Testing and Depyrogenation* (3rd ed.). CRC Press.

Outcomes	➢ Acquire Knowledge on food product analysis ➢ Impart knowledge of preservation technology. ➢ Knowledge on quality analysis of marine food products

Name of the Course Teacher: Dr. T. Sathiamoorthi

Semester II

Core V : Project Work **Maximum Marks: 200**
Code : 517999 **Thesis mark : 150**
Credit : 8 **Viva mark : 50**

Name of Course Teacher	Research supervisor of the M.Phil candidate